Lab program-4:

4. Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next.

```c
lab_4.c > main()
#include<stdio.h>
int main()
{
int bt[20],p[20],wt[20],tat[20],i,j,n,total=0,pos,temp; float
avg_wt,avg_tat;
printf("Enter number of process:");
scanf("%d",&n);
printf("nEnter Burst Time:\n"); for(i=0;i<n;i++)
{
printf("p%d:",i+1);
scanf("%d",&bt[i]);
p[i]=i+1;
}
for(i=0;i<n;i++){
pos=i;
for(j=i+1;j<n;j++)
{
}
if(bt[j]<bt[pos])
pos=j;
temp=bt[i];
bt[i]=bt[pos];
bt[pos]=temp;
temp=p[i];
p[i]=p[pos];
p[pos]=temp;
}
wt[0]=0;
for(i=1;i<n;i++)
{
}
wt[i]=0;
for(j=0;j<i;j++)
wt[i]+=bt[j];
total+=wt[i];
avg_wt=(float)total/n;
total=0;
printf("nProcesst Burst Time tWaiting TimetTurnaround Time\n");
for(i=0;i<n;i++)
{
tat[i]=bt[i]+wt[i];

total+=tat[i];
printf("np%dtt %dtt %dttt%d",p[i],bt[i],wt[i],tat[i]);
}
avg_tat=(float)total/n;
printf("nnAverage Waiting Time=%f",avg_wt);
printf("nAverage Turnaround Time=%fn",avg_tat);
}
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_4.exe
Enter number of process:4
nEnter Burst Time:
p1:7
p2:4
p3:5
p4:2
nProcesst Burst Time tWaiting TimetTurnaround Time
np-1003392952tt -1474298352tt 0ttt-1474298352np2tt 4tt 0ttt4np3tt 5tt -895594496ttt-895594491np4tt 2tt 32760ttt32762nnAverage Waiting Time=-368574592
.000000nAverage Turnaround Time=481276800.000000n
```